

Problem Set 1

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Problem 1

A random variable X takes values $\{1, 2, 4\}$ with probabilities $\{0.2, 0.5, 0.3\}$:

$$P(X = 1) = 0.2, \quad P(X = 2) = 0.5, \quad P(X = 4) = 0.3$$

a)

Compute the expected value $\mathbb{E}[X]$.

The formula for expected value is $\mathbb{E}[X] = \sum x_i P(X = x_i)$

The expected value for X is

$$\mathbb{E}[X] = \sum_{x_i} x_i P(X = x_i) = 1 * 0.2 + 2 * 0.5 + 4 * 0.3 = 2.4$$

b)

Explain what the expected value represents in this context.

The expected value is the weighted sum of all possible outcomes, where the weight of each outcome is its respective probability.

Problem 2

Using the random variable X from Problem 1:

a)

Compute $\mathbb{E}[X^2]$.

$$\mathbb{E}[X^2] = \sum x_i^2 P(X = x_i) = 1 * 0.2 + 4 * 0.5 + 16 * 0.3 = 8.8$$

b)

Compute the variance $Var(X)$.

The formula for variance is $Var(X) = \mathbb{E}[(X - \bar{X})^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

$$\mathbb{E}[X^2] - \mathbb{E}[X]^2 = 8.8 - 2.4^2 = 3.04$$

c)

Compute the standard deviation σ_X .

The mean of X is $1 + 2 + 4/3 = 7/3 = 2.\bar{3}$.

The formula for standard deviation of a sample is:

$$\sigma = \sqrt{\frac{\sum x_i (x_i - \bar{X})^2}{N - 1}}$$

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum x_i (x_i - \bar{X})^2}{N - 1}} \\ &= \sqrt{\frac{(1 - 2.\bar{3})^2 + (2 - 2.\bar{3})^2 + (4 - 2.\bar{3})^2}{2}} \\ &= \sqrt{\frac{-1.\bar{3}^2 + -0.\bar{3}^2 + 1.\bar{6}^2}{2}} \\ &= \sqrt{\frac{-1.\bar{7} + -0.\bar{1} + 2.\bar{7}}{2}} \\ &= \sqrt{\frac{-1.\bar{7} + -0.\bar{1} + 2.\bar{7}}{2}} \\ &= \sqrt{\frac{4.\bar{6}}{2}} \\ &= \sqrt{2.\bar{3}} \\ &\approx 1.528 \end{aligned}$$

d)

Interpret the meaning of the standard deviation.

The standard deviation measures dispersion about a mean.

Approximately 68% of data fall within 1.528 of the mean.

Problem 3

a)

A trader receives \$1 if a stock goes up tomorrow and \$0 otherwise. The probability the stock goes up is $p = 0.6$.

b)

Define the random variable X .

$$P(X) = \begin{cases} 0.4, & X = \$0 \\ 0.6, & X = \$1 \end{cases}$$

c)

Identify the distribution of X .

The distribution of X is discrete, binary. This is a binomial distribution. A binomial distribution with one trial is a Bernoulli distribution.

d)

Compute $\mathbb{E}[X]$, $\text{Var}(X)$, and σ_X .

$$\mathbb{E}[X] = \$0 * 0.4 + \$1 * 0.6 = \$0.60$$

$$\bar{X} = \frac{0 + 1}{2} = 0.5$$

$$\text{Var}_{\text{population}}(X) = \frac{-0.5^2 + 0.5^2}{2} = 0.25$$

$$Var_{sample}(X) = \frac{-0.5^2 + 0.5^2}{1} = 0.5$$

$$\sigma_X^{population} = \sqrt{Var_{population}(X)} = 0.5$$

$$\sigma_X^{sample} = \sqrt{Var_{sample}(X)} \approx 0.707$$

Problem 4

A sales makes $n=5$ independent sales calls, each with probability of success $p = 0.3$.

a)

Define the random variable X .

b)

Identify the distribution of X .

c)

Compute $P(X = 3)$.

d)

Compute $\mathbb{E}[X]$ and $Var(X)$.

Problem 5

0.1 a)

0.2 b)

Problem 6

0.3 a)

0.4 b)

0.5 c)

0.6 d)

Problem 7

0.7 a)

0.8 b)

Problem 8

0.9 a)

0.10 b)

0.11 c)